



Modeling mass and heat transfer in membrane humidifiers for polymer electrolyte membrane fuel cells

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ARTICLE INFO

Keywords:

PEM fuel cells
Membrane
Humidification
Heat and mass transfer
Modeling

ABSTRACT

Water management is essential for the performance and durability of fuel cells in automotive applications. External membrane humidification is the preferred method for this task, as no additional energy or water supply is required. Accurate models of the water transport through the semipermeable membrane can support the design and operation of humidifiers. In this work, an analytical model is proposed that accounts for coupled mass and heat transfer, gas boundary layers, developing flows, and membrane properties at typical operating conditions. The model is validated against experimental data measured with pure Nafion® membranes and an experimental composite membrane. The operating conditions cover a range from 40 to 90 % for the relative humidity, from 1.5 to 2.5 bar for the pressure, and from 340 to 360 K for the temperature. Almost all of the predicted data is within $\pm 30\%$ of the experimental values without the need for any fitting parameters. An analysis of the mass transfer resistances reveals that the membrane and boundary layer resistances both contribute significantly to the total resistance.

1. Introduction

Following discussions on climate change and the formation of global environmental movements, polymer electrolyte membrane (PEM) fuel cells, also known as proton-exchange membrane fuel cells, for automotive applications have gained much attention as an alternative to internal combustion engines in recent years. PEM fuel cells are electrochemical membrane reactors that convert chemical into electrical energy. To maintain their efficiency, the humidity in the fuel cell must be closely managed [1,2]. While excess water in the fuel cell can block the membrane electrode assembly and thus reduce its active area, drying-out of the membrane can cause a drop in proton conductivity and damage the membrane. Both scenarios must be avoided, since they decrease the performance and lifespan of the fuel cell.

To prevent the fuel cell from drying out, the supply air stream can be humidified before entering the cell. There are several different concepts for the humidification of the supply air stream. Examples include bubblers, enthalpy wheels, direct injection, and membrane humidification. The first three methods either induce energy losses, require additional water supply or are very complex to integrate into the fuel cell system.

Membrane humidification avoids these disadvantages and is therefore a promising option for water management in fuel cell applications. [3–5]

Since water is produced in the fuel cell reaction, the exhaust stream exiting the cell is humid. This exhaust stream is passed over a membrane in the humidifier to transfer water selectively to the dry air supply stream. Common module geometries for automotive applications are hollow fiber and flat sheet membrane modules. For both module types, the flow configuration can be either in co-, counter- or cross-current flow. Since the focus of this paper lies on flat sheet membrane modules, the model we will introduce is based on this geometry. In flat sheet membrane modules, the gas streams pass through an array of flow channels on either side of the membrane. Multiple layers of membrane and flow fields are stacked to increase the membrane surface area. In technical modules, cross-current flow is commonly used, since it allows for easier construction.

Viable membranes need to be highly selective with regard to water, thermally stable at the operating temperature of fuel cells, and offer high mechanical and chemical stability. For these reasons, perfluoro-sulfonic acid (PFSA) polymers are widely used both as the polymer electrolyte membrane in the fuel cell stack itself and for membrane hu-

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List of Symbols

a	Thermal diffusivity	m^2/s	ν	Kinematic viscosity	m^2/s
A	Area	m^2	ξ	Relative position	-
c	Molar concentration	kmol/m^3	ρ	Density	kg/m^3
d	Characteristic length	m	σ	Standard deviation	m
$D_{i,j}$	Diffusion coefficient of i in j	m^2/s	τ	Tortuosity	-
$\bar{D}_{i,j}$	Integral diffusion coefficient of i in j	m^2/s	φ	Relative humidity	-
EW	Equivalent weight	kg/kmol	Ψ	Pore size distribution	$1/\text{m}$
f	Sorption isotherm	-	ω	Loading	kg/kg
h	Specific enthalpy	J/kg	Superscripts		
H	Channel height	m	in	Humidifier inlet	
$\dot{H}$	Enthalpy flow	W	out	Humidifier outlet	
j	Mass flux	$\text{kg}/(\text{m}^2 \text{s})$	Subscripts		
L	Channel length	m	air	Air	
$\dot{M}$	Mass flow	kg/s	d	Dry side	
$\bar{M}$	Molar mass	kg/kmol	g	Gaseous	
n	Number of cells	-	H_2O	Water	
$\dot{n}$	Mole flux	$\text{kmol}/(\text{m}^2 \text{s})$	in	Inlet	
p	(Partial) pressure	Pa	i	Stream i	
P	Permeability	$\text{kmol}/(\text{s m Pa})$	j	Mass flux	
Q	Permeance	GPU	Kn	Knudsen	
$\dot{Q}$	Heat flow	W	l	Liquid	
T	Temperature	K	log	Logarithmic	
r	Pore radius	m	m	Membrane	
R	Mass transfer resistance	m/s	mean	Mean	
$\bar{R}$	Universal gas constant	$\text{J}/(\text{kmol K})$	P	Poiseuille	
u	Velocity	m/s	s	Saturation	
$\dot{V}_N$	Volumetric flow rate at standard conditions	m^3/s	sl	Support layer	
W	Channel width	m	total	total	
x	Coordinate along channel length	m	w	Wet side	
z	Coordinate perpendicular to membrane plane	m	wall	Wall	
Greek Symbols					
α	Convective heat transfer coefficient	$\text{W}/(\text{m}^2 \text{K})$	Dimensionless numbers		
β	Convective mass transfer coefficient	m/s	Nu	Nusselt number, $Nu = \alpha d / \lambda$	
δ	Thickness	m	Pr	Prandtl number, $Pr = \nu / \alpha$	
Δ	Difference	-	Re	Reynolds number, $Re = d u / \nu$	
ϵ	Porosity	-	Sc	Schmidt number, $Sc = \nu / D$	
λ	Thermal conductivity	$\text{W}/(\text{m K})$	Sh	Sherwood number, $Sh = \beta d / D$	
Λ	Water content	kmol/kmol			

modifiers. Nafion[®] membranes produced by DuPont de Nemours are the most prominent membranes of this type. Modern membrane design is moving towards composite membranes comprising very thin PFSA layers reinforced by support layers of different material to maximize water transport.

To predict the water transport in membrane humidifiers under varying operating conditions, numerous models have been published to describe mass and heat transfer. The models can be grouped into shortcut models, analytical models and computational fluid dynamic (CFD) models.

In shortcut models, the mass and heat transfer is described using correlations for the overall effectiveness and for the number of transfer units [6,7]. While they are quick and easy to use, they lack a detailed analysis of physical phenomena.

CFD models are capable to give detailed insights, but they are the most complex and most computationally intensive type of models. Aspects of research using CFD modeling are mostly mass and heat transfer as well as geometric design [8–16].

Analytical models provide a good balance between shortcut and CFD models with respect to implementation effort, computational resources, and detailed insights. The basis of these models is local mass and heat balances based on different assumptions regarding the mass and heat

transfer. Examples for air-to-air membrane humidifier models can be found in [17–26]. Differences between these models include the coupling of mass and heat transfer, the treatment of gas boundary layers, and the dependence of properties on the process parameters.

Zhang and Jiang [17] use their model to analyze how different membrane regions of a cross-current humidifier contribute to mass and heat transfer. They find that the inlet regions contribute the most to the water transfer. They conclude that counter-current is preferable to cross-current configuration. Chen and Peng [18] propose a dynamic counter-current model. They use steady-state results for design purposes and perform a dynamic behavior analysis. They come to the conclusion that the counter-current configuration is also better than a co-current configuration using a dynamic model. Their steady-state results show how varying channel geometries effect the heat and mass transfer due to changes in the membrane area. The effects of geometric parameters on the water transfer are also confirmed by Kang et al. [19] and Yu et al. [20]. They also investigate the influence of operating conditions such as relative humidity, temperature, pressure and flow rate based on their models for a shell-and-tube humidifier and a flat sheet membrane humidifier, respectively. They conclude that the relative humidity at the inlets has a strong influence on the water transfer as it directly impacts the gradient.

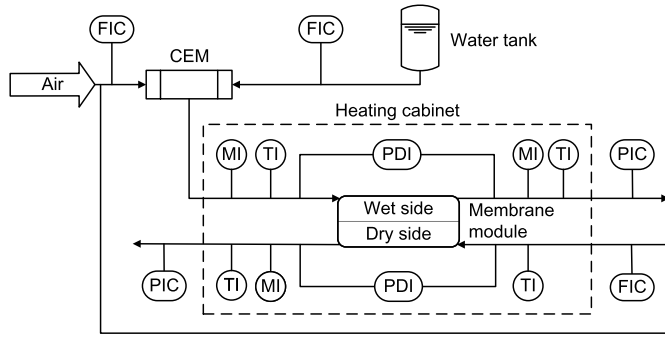


Fig. 1. Simplified piping and instrumentation diagram of the experimental setup.

Sabharwal et al. [21] and Bhatia et al. [22] derive differential equation systems for cross- and counter-current humidifiers, of which the latter is solved analytically. They perform a thorough analysis on various geometric and operating parameters. Min and Duan [23] apply their cross-current model for air conditioning purposes at different weather scenarios and find that the water transfer direction can vary across the membrane depending on the conditions. A cross-counter-flow enthalpy exchanger has been modeled by Koester et al. [24]. Based on a sensitivity analysis, they conclude that mass transfer depends on both gas boundary layer and membrane resistance. Yun et al. [25] propose a dynamic one-dimensional model with which they investigate the effects of air acceleration and sinusoidal air flow rates. Engarnevis et al. [26] develop a cross-current model for air-to-air energy recovery ventilators with asymmetric composite membranes. They show the importance of membrane orientation and resolving the membrane permeability locally.

In this publication, we propose an application-driven analytical model for mass and heat transfer in PFSA-based flat sheet membrane humidifiers. Prior measurements [27–29] show that the operating conditions such as relative humidity, temperature, pressure, and flow rate have a major impact on the water transfer characteristics. The proposed model is capable to capture these influences by describing the individual transport steps with physical and empirical equations. This results in a first-principle model without the need for fitting parameters. The focus of this model is the phenomena that were found to have a significant impact in technical applications, including gas boundary layers, developing flows in entry regions and coupled mass and heat transfer. Thus, highly complex models for the sorption and diffusion of water in PFSA membranes are not incorporated, as many of the material properties in state-of-the-art membranes are not disclosed by manufacturers. The model in this paper is compared to experimental data resembling conditions in automotive applications. The experimental data includes pure Nafion® membranes and a composite membrane.

2. Experimental setup

In this work, experimental data for the water transport of Nafion® 211, 212, and 115 membranes as well as an experimental composite membrane by W. L. Gore & Associates are used for model validation. The experimental setup is described in detail by Wolfenstetter et al. [29]. Since we will show additional data, this section gives a short summary of the experimental setup.

The experimental setup shown in Fig. 1 is used to measure the water transport through a membrane sample under typical operating conditions of a membrane humidifier in fuel cell systems. Compressed air is preconditioned to remove particles, oil, and water. The conditioned air stream is split into two streams, leading to the dry and wet side of the membrane module, respectively. The stream on the dry side enters the membrane module directly and simulates the air supply stream into the fuel cell. It is kept at a constant relative humidity of $\varphi_d^{\text{in}} \leq 0.1\%$

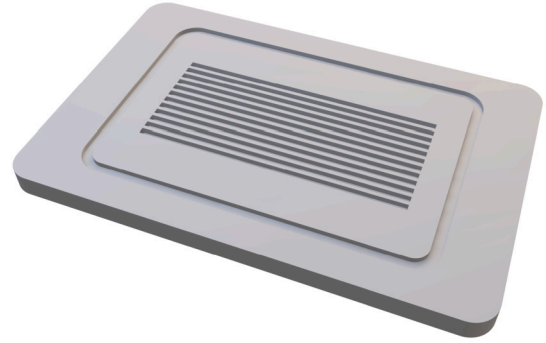


Fig. 2. CAD model of the flow field.

Table 1

Varied parameters of the measurement series.

Series	Wet inlet humidity φ_w^{in} in %	Temperature T in K	Pressure p in bar
φ	40, 50, ..., 90	350	2.5
T	85	340, 350, 360	2.5
p	85	350	1.5, 1.75, ..., 2.5

throughout all experiments. Deionized water is added to the stream on the wet side with a controlled evaporator mixer (CEM) before entering the membrane module. This wet stream represents the exhaust stream from the fuel cell. The humidity of the wet stream at the inlet is controlled. At the inlet of the wet side and the outlets of the wet and dry side, the dew points are measured with a high-precision chilled-mirror hygrometer (Michell Instruments S8000 Remote) with an accuracy of $\pm 0.1^\circ\text{C}$. From the measured dew points and the gas flow rates, the mass flux of water through the membrane sample can be calculated. Additionally, for each gas stream, the pressure drop across the membrane module and the temperatures at the in- and outlets are recorded.

The membrane module consists of two halves, between which the membrane sample is sandwiched. Both module halves contain a flow field as depicted in Fig. 2. The geometry of the flow field has a strong influence on the water transfer in a membrane humidifier [30,31]. The used flow fields consist of 12 straight, rectangular channels with the width $W = 1.25\text{ mm}$, the height $H = 1.10\text{ mm}$, and the length $L = 54\text{ mm}$. This simple geometry design is easy to manufacture and enables the description of the flow conditions in the flow field using correlations available in the literature. The flow field is used for the Nafion® membranes and the experimental composite membrane by W. L. Gore & Associates. A gasket ensures that water can pass from the wet to the dry side only through the membrane. Before assembly, Nafion® membranes are pretreated with a standard cleaning protocol, as described in [29].

For each membrane, the water transport is measured for different values of the relative humidity φ_w^{in} of the wet inlet stream, the temperature T , and the absolute pressure p at the inlet. One parameter is varied at a time while the other two parameters are kept constant. Table 1 summarizes the measurement conditions. For each series, the volumetric flow rate $\dot{V}_N$ at standard conditions is varied between 5 to 27.5 L/min for the wet and dry stream, respectively. The volumetric flow rate is identical for both sides. The flow rates correspond to dry inlet Reynolds numbers Re between 350 to 2050 in a channel for the used flow field. All measurements are conducted with the module in counter-current configuration.

3. Mass transfer through the membrane

Common membrane types can be categorized into porous and dense membranes. In membrane humidifiers, dense membranes that are selective for water vapor are predominantly used. The mass transfer mecha-

nism in dense membranes is described with the solution-diffusion model [32]. Thus, the following steps for the mass transport from the wet bulk stream through the membrane into the dry bulk stream are considered:

1. Transport through the wet stream boundary layer
2. Sorption at the PFSA surface
3. Diffusion through the PFSA layer
4. Desorption at the PFSA surface
5. Transport through the dry stream boundary layer

The steps are described in the following sections. In case of composite membranes, one must consider additional steps for the support layers.

3.1. Transport through boundary layers

The effect of concentration polarization occurs in membrane humidification processes, because water is selectively transferred through the membrane, while other components in the gas mixture remain on their respective side of the membrane. This leads to a depletion of water in the proximity of the membrane on one side and an enrichment of water on the other side. The transport between the bulk phase of a gas stream and the membrane surface is characterized by a combination of diffusion and advection. For the thermodynamically correct representation of this convective mass transfer, the boundary layer must be resolved and the gradient in the chemical potential in the boundary layer must be determined. For the purpose of an application-driven analytical model, a more straightforward approach is desirable. A well-established approach in process engineering is the use of the concentration difference $\Delta c_{\text{H}_2\text{O}}$ as the driving force in combination with a convective mass transfer coefficient β to obtain the molar flux

$$\dot{n} = \beta \Delta c_{\text{H}_2\text{O}}. \quad (1)$$

By applying the ideal gas law, this can be converted to

$$j = \beta \frac{\bar{M}_{\text{H}_2\text{O}}}{\bar{R}T} \Delta p_{\text{H}_2\text{O}}, \quad (2)$$

with j being the mass flux, $\bar{M}_{\text{H}_2\text{O}}$ the molar mass of water, $\bar{R}$ the universal gas constant, T the temperature and $\Delta p_{\text{H}_2\text{O}}$ the water vapor partial pressure difference between the bulk phase and the membrane surface.

The mass transfer coefficient β can be obtained using correlations for the Sherwood number Sh . Sherwood correlations are usually functions of the Reynolds number Re and the Prandtl number Pr . We assume analogy of heat and mass transfer and convert the Nusselt correlation of Muzychka and Yovanovich [33] into a Sherwood correlation by replacing the Nusselt number Nu with the Sherwood number Sh and the Prandtl number Pr with the Schmidt number Sc . Zhang et al. [34] have shown that the thermal and hydraulic entry regions can amount to a significant part of the flow field and that the assumption of fully developed flows can lead to severe errors. Therefore, the correlation of Muzychka and Yovanovich [33] for local Nusselt numbers in non-circular ducts is used to account for entry effects. The assumptions of uniform wall temperature (UWT) or uniform wall flux (UWF) are commonly made in heat transfer applications. It is understood, as described in detail by Zhang et al. [34], that neither of these idealizations is true for coupled heat and mass transfer in membrane applications. Similar to Engarnevis et al. [26], we use the correlation for UWT of Muzychka and Yovanovich [33]. It should be noted that in our setup only one of four walls in the channel is active in heat and mass transfer. The correlation of Muzychka and Yovanovich [33] does not consider this. However, we decided to use it as it provides local Nusselt numbers for a rectangular duct and we are not aware of a correlation that additionally accounts for inactive walls.

For the system of water vapor and air, it can be convenient to express the partial pressure $p_{\text{H}_2\text{O}}$ in terms of the relative humidity φ , which relates the partial pressure $p_{\text{H}_2\text{O}}$ of water vapor to its saturation pressure p_s :

$$\varphi = \frac{p_{\text{H}_2\text{O}}}{p_s(T)}. \quad (3)$$

3.2. Sorption

Empirical sorption isotherms correlate the concentration of water in the membrane with the relative humidity in the gas phase. Usually, the concentration in the membrane is expressed as the water content Λ , which is a measure of the moles of water molecules per moles of sulfonic acid groups in the PFSA membranes. The concentration of water $c_{\text{H}_2\text{O}}$ is related to the water content Λ by

$$\Lambda = \frac{c_{\text{H}_2\text{O}} EW}{\rho_m}, \quad (4)$$

where ρ_m is the density of the dry membrane and EW is the equivalent weight, which is defined as the mass of dry Nafion[®] per mole of sulfonic acid groups. Empirical sorption isotherms for Nafion[®] and corresponding correlations have been determined by many different researchers. The dependency of the water content Λ on the relative humidity φ was fitted based on different approaches, such as polynomials [35–39], BET isotherms [40,41], hydration shell models [42–44], New-Dual-Mode-Sorption isotherms [45–47], and others [48–50]. For this work, the isotherm of Mittelsteadt and Liu [37] was chosen as it is valid for conditions similar to the conditions stated in Section 2:

$$\Lambda = \left[1 + 0.2325 \varphi^2 \left(\frac{\left[\frac{T}{^\circ\text{C}} \right] - 30}{30} \right) \right] \cdot [14.22 \varphi^3 - 18.92 \varphi^2 + 13.41 \varphi]. \quad (5)$$

While Nafion[®] has been used as a polymer electrolyte membrane for a long time, there is still an ongoing debate about the existence of an interfacial resistance and the contributing phenomena [51]. Adachi et al. [52,53] explain their results with the existence of an interfacial resistance and attribute this to phenomena occurring at the membrane interface. According to their results, the interfacial resistance becomes more important with decreasing membrane thickness. Majsztrik et al. [54,55] ascribe the interfacial transfer resistance to the gas boundary layer and the phase change at the membrane surface. According to Zhao et al. [56], the boundary layer resistance for the measurements performed by Majsztrik et al. [55] is negligible due to the high gas velocities in the experimental setup. On the contrary, Caulk et al. [57] summarize resistances, not stemming from the diffusion in the membrane, as device resistances. As the name suggests, such resistances depend on the experimental setup. For example, boundary layer resistances are present even at high gas velocities. According to Caulk et al. [57], device resistances become even more significant with decreasing membrane thickness due to decreasing diffusion resistance of the membrane. Thus, it is not easy to distinguish between resistances due to phenomena at the membrane surface and the device resistance. In this work, we assume the boundary layer resistances to be dominating over possibly existing surface resistances for our experimental setup.

3.3. Diffusion in PFSA membranes

Water transport in PFSA membranes can take place via various transport mechanisms. These differ in the governing driving force and are electro-osmosis, thermo-osmosis, hydraulic transport, and diffusion. Depending on the conditions of the system, one or more transport mechanisms can contribute to the overall water transport through the membrane. [58] An approach on how water transport mechanisms in PFSA membranes can be coupled can be found for example in the work of Rangel-Cárdenas and Koper [59] and Kreuer et al. [60]. Since the focus of this work lies on membrane humidifiers and no electric field is present, electro-osmosis is not considered. As the membrane module in

our setup is placed in a heating cabinet, only small temperature gradients between the wet and dry side of the membrane are expected. Therefore, water transport through thermo-osmosis is assumed to be negligible. The hydraulic transport induced by a gradient in pressure is also not considered since the same operating pressure is used for the wet and dry side in our setup. Following these explanations, we assume that the water transport through the membrane is governed by diffusion.

The driving force for the diffusion of a component is the gradient in its chemical potential. Wolfenstetter et al. [29] showed that the gradient in the chemical potential can be simplified into a gradient in the concentration, resulting in Fick's first law:

$$j = -D_{\text{H}_2\text{O},m} \bar{M}_{\text{H}_2\text{O}} \frac{dc_{\text{H}_2\text{O}}}{dz}. \quad (6)$$

In Eq. (6), $D_{\text{H}_2\text{O},m}$ is the diffusion coefficient of water in the membrane. The coordinate z is the direction perpendicular to the membrane plane. Integration of Eq. (6) over the membrane thickness δ_m and replacing the concentration of water $c_{\text{H}_2\text{O}}$ with the water content Λ as defined in Eq. (4) yields:

$$j = \frac{\bar{D}_{\text{H}_2\text{O},m}}{\delta_m} \frac{\bar{M}_{\text{H}_2\text{O}} \varrho_m}{EW} (\Lambda_w - \Lambda_d). \quad (7)$$

The quantities Λ_w and Λ_d represent the water content at the wet and dry membrane surfaces, respectively. Since the diffusion coefficient $D_{\text{H}_2\text{O},m}$ of water in PFSA membranes is a strong function of water content, the average diffusion coefficient $\bar{D}_{\text{H}_2\text{O},m}$ is determined by integration from the water content on the dry side to the water content on the wet side:

$$\bar{D}_{\text{H}_2\text{O},m} = \frac{1}{\Lambda_w - \Lambda_d} \int_{\Lambda_d}^{\Lambda_w} D_{\text{H}_2\text{O},m}(\Lambda) d\Lambda. \quad (8)$$

Numerous authors have presented measurements and correlations for the diffusion coefficient $D_{\text{H}_2\text{O},m}$, which vary by an order of magnitude. Kusoglu and Weber [58] state that the disparity between the different models is believed to stem from different methods of measurements and different methods for deriving the correlations from the experimental data. Nguyen [61] and Fuller [62] present a linear dependency of the diffusion coefficient on water content, while Motupally et al. [63] and Dutta et al. [64] found a piecewise fit to the experimental data of Springer et al. [35] with a pronounced local maximum. Olesen et al. [65,66] have shown that this maximum is a mathematical artifact from the derivation of a concentration dependent diffusivity from the experimental data, and presented a continuous correlation for the diffusion coefficient. Following this argumentation, the correlation by Olesen [66] is used:

$$\left[\frac{D_{\text{H}_2\text{O},m}}{\text{m}^2/\text{s}} \right] = 5.39 \cdot 10^{-6} (1 + 0.0027 \Lambda^2) \cdot \left[1 + \tanh \left(\frac{\Lambda - 2.6225}{0.8758} \right) \right] \exp \left(\frac{-3343}{\left[\frac{T}{\text{K}} \right]} \right). \quad (9)$$

3.4. Transport through the support layer

To maximize water transport, modern composite membranes use very thin selective layers. To compensate the loss of mechanical strength, support layers of a different material can be laminated to one or both sides of the membrane. In this work, an experimental composite membrane manufactured by W. L. Gore & Associates with expanded polytetrafluoroethylene (ePTFE) on both sides of the selective layer is modeled. The support layers are treated as a porous medium and the mass transfer through the pores can be described by the Dusty-Gas-Model. The Dusty-Gas-Model considers molecular diffusion and

Knudsen diffusion as mass transfer resistances in series parallel to viscous Poiseuille flow [26]. The mass flux j in this model is defined as [67]:

$$j = \frac{\epsilon}{\tau \delta_{sl}} \frac{\int_0^\infty \Psi(r) \left[\left(D_{Kn}^{-1} + D_{\text{H}_2\text{O},\text{air}}^{-1} \right)^{-1} + D_p \right] r^2 dr}{\int_0^\infty \Psi(r) r^2 dr} \Delta p_{\text{H}_2\text{O}}. \quad (10)$$

In Eq. (10), the flux j is dependent on the porosity ϵ and the tortuosity τ of the support layer, the support layer thickness δ_{sl} , and the pore radius r . D_{Kn} , $D_{\text{H}_2\text{O},\text{air}}$, and D_p are the diffusion coefficients of Knudsen diffusion, molecular diffusion and Poiseuille flow, respectively. For the pore size distribution $\Psi(r)$, a normal distribution is assumed:

$$\Psi(r) = \frac{1}{\sqrt{2\pi}\sigma} \exp \left[-\frac{1}{2} \left(\frac{r - r_{\text{mean}}}{\sigma} \right)^2 \right]. \quad (11)$$

r_{mean} is the mean pore radius of the support layer. Based on Min and Hu [67], we assume a standard deviation of $\sigma = 0.3 r_{\text{mean}}$ for this material. The Knudsen diffusion coefficient is dependent on the pore radius and is calculated by

$$D_{Kn} = \frac{2}{3} r \sqrt{\frac{8}{\pi} \frac{\bar{R} T}{\bar{M}_{\text{H}_2\text{O}}}}. \quad (12)$$

The molecular diffusion coefficient $D_{\text{H}_2\text{O},\text{air}}$ of water vapor in air is calculated using the Chapman-Enskog model [68]. As described by Engarnevis et al. [26], it is assumed that there is no pressure gradient in the support layers, as the selective layer is a dense polymer. Furthermore, the experiments in this work are conducted with equal pressures on both sides of the membrane. Therefore, no viscous flow in the pores is assumed and the calculation of the flux reduces to

$$j = \frac{D_{sl}}{\delta_{sl}} \Delta p_{\text{H}_2\text{O}} \quad (13)$$

with

$$D_{sl} = \frac{\epsilon}{\tau} \frac{\int_0^\infty \Psi(r) \left(D_{Kn}^{-1} + D_{\text{H}_2\text{O},\text{air}}^{-1} \right)^{-1} r^2 dr}{\int_0^\infty \Psi(r) r^2 dr}. \quad (14)$$

4. Model description

The presented model is similar to the one described by Engarnevis et al. [26] and is adapted to PFSA and PFSA-based membranes. The flow field is discretized into n cells along the channel length seen in Fig. 2. Each cell comprises one segment each of the wet and dry flow channel that are separated by a membrane. The steps described in Section 3 are modeled in each cell. The basic modeling assumptions are:

- It is not necessary to consider all channels of the module separately. Since they behave the same, it is sufficient to consider only one pair of wet and dry channels. This implicitly assumes that feed streams are distributed evenly among the channels.
- The humidifier is in steady state.
- Inside a cell segment, the considered variables are uniformly distributed in space. The air streams leave the cell with these conditions. This corresponds to the assumption of an ideally stirred tank in each cell segment.
- Diffusion is negligible compared to advective transport along the flow direction in the bulk streams.
- The membrane is selectively permeable for water. The air mass flows themselves remain constant within a channel.
- Sorption equilibrium is reached at the air-membrane-interface.

First, we describe the model for pure PFSA membranes. Afterwards, the necessary modifications to account for composite membranes are presented.

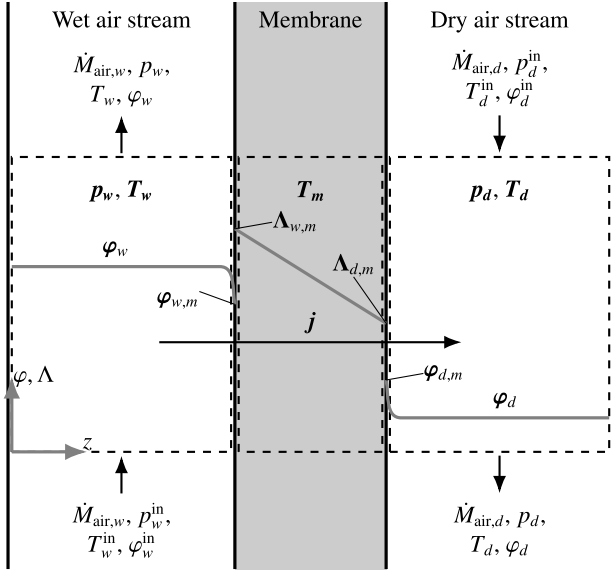


Fig. 3. Schematic of the mass transfer in discretized flow channels. Unknowns are highlighted in bold.

4.1. PFSA membranes

Based on the assumptions listed above, Fig. 3 shows a sketch of the model for PFSA membranes, considering a wet air stream (index w), a membrane (index m) and a dry air stream (index d). The dashed lines indicate the system boundaries. The states of the air streams are determined by their air mass flows $\dot{M}_{\text{air}}$, pressure p , temperature T , and relative humidity ϕ , respectively. The membrane has the temperature T_m and different water contents Λ on the wet and dry side. Due to a partial pressure gradient, water is transferred from the wet bulk stream through the membrane to the dry bulk stream. This results in the water mass flux j . The condition of one cell can be fully determined by the twelve unknowns highlighted in bold in Fig. 3. The governing equations are mass, energy, and momentum balances as well as mass transfer equations. They are introduced below.

For both the wet and the dry side, water mass balances must hold. For modeling purposes, it is convenient to express the water mass flow via the vapor loading ω and the constant air mass flow $\dot{M}_{\text{air}}$:

$$\dot{M}_{\text{air},w} \omega_w^{\text{in}} = j A_m + \dot{M}_{\text{air},w} \omega_w \quad (15)$$

$$\dot{M}_{\text{air},d} \omega_d^{\text{in}} + j A_m = \dot{M}_{\text{air},d} \omega_d \quad (16)$$

A_m is the membrane surface area at each side within a cell given by $A_m = L \cdot W/n$. Based on Dalton's law, the relationship between the vapor loading ω and the relative humidity ϕ is

$$\omega = \frac{\bar{M}_{\text{H}_2\text{O}}}{\bar{M}_{\text{air}}} \frac{p_s(T) \phi}{p - p_s(T) \phi}, \quad (17)$$

with p being the total pressure.

Due to the steady state, the mass flux j is the same for all mass transfer steps, i.e. through the wet stream boundary layer

$$j = \beta_w \frac{\bar{M}_{\text{H}_2\text{O}}}{\bar{R} T_w} \left(p_s(T_w) \phi_w - p_s(T_m) \phi_{w,m} \right), \quad (18)$$

through the membrane

$$j = \frac{\bar{D}_{\text{H}_2\text{O},m}}{\delta_m} \frac{\bar{M}_{\text{H}_2\text{O}} \phi_m}{EW} (\Lambda_w - \Lambda_d), \quad (19)$$

and through the dry stream boundary layer

$$j = \beta_d \frac{\bar{M}_{\text{H}_2\text{O}}}{\bar{R} T_d} \left(p_s(T_m) \phi_{d,m} - p_s(T_d) \phi_d \right). \quad (20)$$

At the air-membrane-interfaces, sorption equilibrium is reached:

$$\Lambda_w = f(\phi_{w,m}, T_m) \quad (21)$$

$$\Lambda_d = f(\phi_{d,m}, T_m). \quad (22)$$

Here, f is the sorption isotherm given in Eq. (5).

Even though the heating cabinet in the experimental setup provides a nearly isothermal environment, the energy balance is used to capture sorption effects and to access the membrane temperature T_m , which is important for the sorption behavior, see Eqs. (21) and (22). We consider the convective heat transfer ($\dot{Q}_{w,m}$, $\dot{Q}_{d,m}$) between the air streams and the membrane, the convective heat transfer ($\dot{Q}_{\text{wall},w}$, $\dot{Q}_{\text{wall},d}$) between the channel walls and the air streams, the advective enthalpy transport ($\dot{H}_w$, $\dot{H}_d$, $\dot{H}_w^{\text{in}}$, $\dot{H}_d^{\text{in}}$) induced by the air streams, and the enthalpy transport ($\dot{H}_{w,j}$, $\dot{H}_{d,j}$) induced by the water flux through the membrane:

$$\dot{Q}_{\text{wall},w} + \dot{H}_w^{\text{in}} = \dot{H}_w + \dot{Q}_{w,m} + \dot{H}_{w,j} \quad (23)$$

$$\dot{Q}_{w,m} + \dot{H}_{w,j} = \dot{Q}_{d,m} + \dot{H}_{d,j} \quad (24)$$

$$\dot{Q}_{\text{wall},d} + \dot{H}_d^{\text{in}} + \dot{Q}_{d,m} + \dot{H}_{d,j} = \dot{H}_d. \quad (25)$$

The convective heat flow between an air stream i and the membrane is

$$\dot{Q}_{i,m} = \alpha_i A_m (T_i - T_m), \quad (26)$$

with the heat transfer coefficient α_i . Similar, the convective heat flow between the channel walls and the air streams is

$$\dot{Q}_{\text{wall},i} = \alpha_i A_{\text{wall}} (T_{\text{wall}} - T_i), \quad (27)$$

with the cell wall surface $A_{\text{wall}} = L/n \cdot (W + 2H)$. The advective enthalpy flows $\dot{H}_i$ are given by

$$\dot{H}_i = \dot{M}_{\text{air},i} (1 + \omega_i) h_i. \quad (28)$$

Here, h_i is the specific enthalpy of stream i . The water flux j enters the membrane system boundary as water vapor (index g), whereas it leaves it as sorbed water. The enthalpy of sorbed water is assumed to be the same as that of liquid water (index l):

$$\dot{H}_{w,j} = j A_m h_{\text{H}_2\text{O},g}(T_w) \quad (29)$$

$$\dot{H}_{d,j} = j A_m h_{\text{H}_2\text{O},l}(T_m). \quad (30)$$

Note that the sorption enthalpy is considered implicitly in Eq. (24) with the definition of $\dot{H}_{w,j}$ and $\dot{H}_{d,j}$.

In the momentum balances, the change of momentum due to the transferred water is assumed to be negligible compared to the pressure drop Δp within a cell:

$$p_w = p_w^{\text{in}} - \Delta p_w \quad (31)$$

$$p_d = p_d^{\text{in}} - \Delta p_d. \quad (32)$$

The pressure drops are calculated according to [69].

Equations (15), (16), (18) to (25), (31) and (32) set an equation system for the determination of the 12 unknowns. The equation system is solved using MATLAB's nonlinear optimizer `fmincon`.

One pair of channels is modeled by a one-dimensional array of cells. The inlet streams of one cell are set equal to the streams in the respective upstream cells. Depending on how the streams between the cells are connected, co-current or counter-current can be modeled. The representation of cross-current is also possible with a two-dimensional array of cells, but is not considered in this work.

4.2. Composite membranes

In the case of composite membranes with a support layer on each side of the PFSA layer, two additional unknowns have to be determined.

Table 2
Properties of the investigated membranes.

Membrane	δ_m in μm	EW in $\frac{\text{kg}}{\text{kmol}}$	ρ_m in $\frac{\text{kg}}{\text{m}^3}$
Nafion® 211	25.4	1000	1968.5
Nafion® 212	50.8	1000	1968.5
Nafion® 115	127.0	1100	1968.5
Gore composite	< 10	800	2000.0

Table 3
Properties of the support layer for the Gore composite membrane.

Thickness δ_{sl}	10 μm
Porosity ϵ	0.8
Tortuosity τ	1.25
Mean pore radius r_{mean}	0.125 μm

These are the relative humidities $\varphi_{w,sl}$ and $\varphi_{d,sl}$ at the interfaces between the support layers and the wet and dry air stream, respectively. Therefore, Eqs. (18) to (20) are expanded by two equations and replaced with

$$j = \beta_w \frac{\bar{M}_{\text{H}_2\text{O}}}{\bar{R} T_w} (p_s(T_w) \varphi_w - p_s(T_m) \varphi_{w,sl}) \quad (33)$$

$$j = \frac{D_{w,sl}}{\delta_{sl}} \frac{\bar{M}_{\text{H}_2\text{O}} p_s(T_m)}{\bar{R} T_m} (\varphi_{w,sl} - \varphi_{w,m}) \quad (34)$$

$$j = \frac{\bar{D}_{\text{H}_2\text{O},m}}{\delta_m} \frac{\bar{M}_{\text{H}_2\text{O}} \varphi_m}{EW} (\Lambda_w - \Lambda_d) \quad (35)$$

$$j = \frac{D_{d,sl}}{\delta_{sl}} \frac{\bar{M}_{\text{H}_2\text{O}} p_s(T_m)}{\bar{R} T_m} (\varphi_{d,m} - \varphi_{d,sl}) \quad (36)$$

$$j = \beta_d \frac{\bar{M}_{\text{H}_2\text{O}} p_s}{\bar{R} T_d} (p_s(T_m) \varphi_{d,sl} - p_s(T_d) \varphi_d) \quad (37)$$

Equations (34) and (36) account for the mass transfer through the support layer.

5. Results and discussion

In this section, the model is validated using experimental data of pure Nafion® membranes of different thicknesses and an experimental composite membrane manufactured by W. L. Gore & Associates, which will be referred to as Gore composite membrane in the further course. In the experiments, the relative humidity at the wet inlet, temperature and pressure were varied in sequence as specified in Table 1. All data were obtained using the experimental setup described in Section 2. Some of the data presented in this paper is already published in [29]. For the sake of conciseness, we present and discuss the yet unpublished experimental data for Nafion® 212 in detail and use it for validation of the model. For the other membranes, parity plots are used to evaluate the model performance. The properties of the investigated Nafion® membranes as well as the Gore composite membrane are summarized in Tables 2 and 3. The tortuosity τ and porosity ϵ are taken from Li and Sirkar [70].

For the computation of the model results, a resolution of $n = 100$ cells was found to be sufficient by a grid independence study. In accordance to [29], the results are reported as the permeance Q . This is the ratio between permeability P and the membrane thickness δ_m and can be defined as

$$Q = \frac{P}{\delta_m} = \frac{\dot{V}_N}{A_m \Delta p_{\text{H}_2\text{O},\log}}, \quad (38)$$

where $\dot{V}_N$ is the volumetric flow rate of water vapor at standard conditions and $\Delta p_{\text{H}_2\text{O},\log}$ is the logarithmic mean difference in water partial

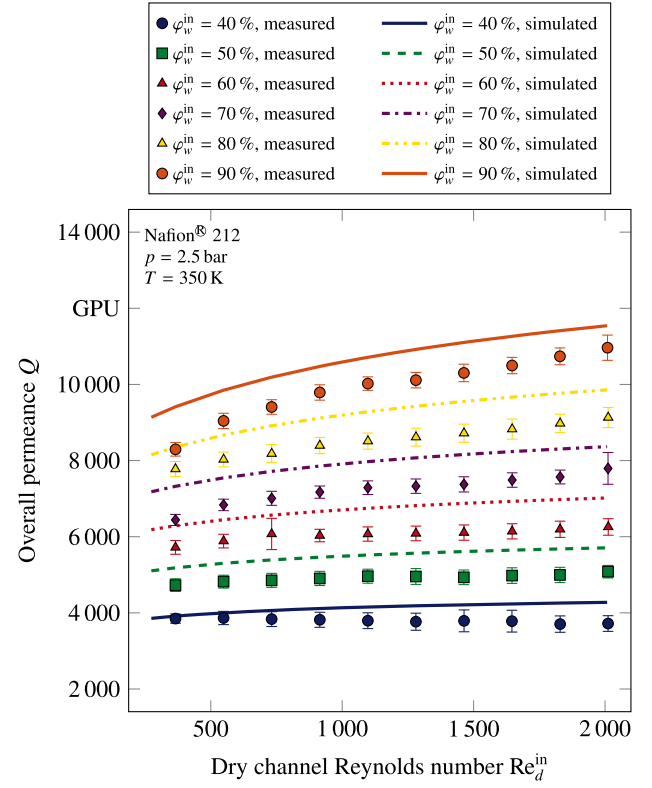


Fig. 4. Permeance Q of Nafion® 212 as a function of the dry channel Reynolds number Re_d^{in} for varying relative humidity φ_w^{in} .

pressure. It can be calculated using inlet and outlet partial water pressures. For counter-current it is defined as

$$\Delta p_{\text{H}_2\text{O},\log} = \frac{(p_{\text{H}_2\text{O},w}^{\text{in}} - p_{\text{H}_2\text{O},d}^{\text{out}}) - (p_{\text{H}_2\text{O},w}^{\text{out}} - p_{\text{H}_2\text{O},d}^{\text{in}})}{\ln \left(\frac{p_{\text{H}_2\text{O},w}^{\text{in}} - p_{\text{H}_2\text{O},d}^{\text{out}}}{p_{\text{H}_2\text{O},w}^{\text{out}} - p_{\text{H}_2\text{O},d}^{\text{in}}} \right)}. \quad (39)$$

Essentially, the permeance is a molar flux related to the driving force. The permeance Q is usually given in gas permeation units, defined as the volumetric flow rate at standard conditions per area and pressure difference:

$$1 \text{ GPU} = 10^{-6} \frac{\text{cm}^3}{\text{s cm}^2 \text{ cmHg}}. \quad (40)$$

5.1. Nafion® membranes

The model results for the permeance Q are compared to the experimental values for different measurement series of Nafion® 212 in Figs. 4 to 6.

Impact of relative humidity In Fig. 4, the permeance of Nafion® 212 is shown over the Reynolds number at the dry gas channel inlet for a pressure of $p = 2.5$ bar, a temperature of $T = 350$ K and varying relative humidities at the inlet of the wet gas channel ranging from 40 to 90%.

Generally, permeance of the membrane increases with higher values of relative humidity and rising Reynolds numbers. The increase of permeance with relative humidity in the wet gas stream is much more pronounced than the influence of the Reynolds number. With rising relative humidity, the water content in the membrane increases, leading to a higher diffusion coefficient of the membrane.

The experimental setup does not allow to resolve the concentration boundary layers adjacent to the membrane. Instead, an overall water transfer through the membrane and the boundary layers is measured. Since the boundary layers are strongly dependent on the flow

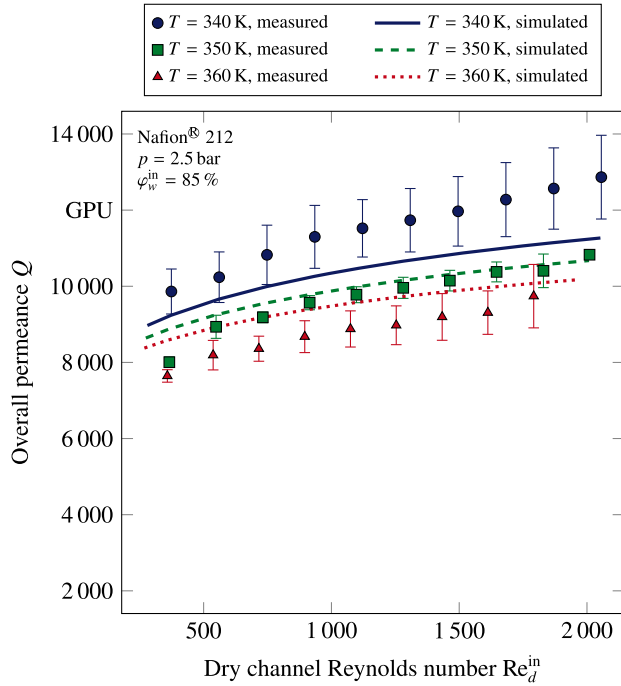


Fig. 5. Permeance Q of Nafion® 212 as a function of the dry channel Reynolds number Re_d^{in} for varying temperature T .

conditions, the experimental permeance exhibits a dependence on the Reynolds number. With higher Reynolds numbers, the boundary layer is thinner, resulting in a higher mass transfer coefficient in the gas phase.

Furthermore, the influence of the Reynolds number on the permeance decreases with lower relative humidities in the wet channel. For lower relative humidities, the diffusion coefficient becomes more dominating. The effects of relative humidity and the Reynolds number are described well by the presented model and will be discussed in Section 5.3.

Impact of temperature A change in temperature affects the water transport in multiple ways. Higher temperatures increase the diffusion coefficients in both the gas phase and the membrane. In addition, the equilibrium water content of the membrane is also higher at higher temperatures. The increased diffusion coefficients in the membrane and the gas phase promote water transport through the membrane, but with regard to the permeance in Fig. 5, the opposite is observed. By its definition in Eq. (38), permeance is the mass flux of water in relation to the logarithmic water vapor partial pressure difference. The saturation pressure of water increases strongly with increasing temperature. At constant relative humidities, this results in increasing partial pressure of water according to Eq. (3). Therefore, the driving force increases, which outweighs the higher mass flux, leading to lower overall permeance with rising temperatures. The temperature dependency is represented in the model results. However, the dependency is underestimated in the model. The identification of reasons is challenging since the temperature effects on the diffusion coefficients and the sorption behavior are interdependent and cannot be separated.

Impact of pressure Fig. 6 shows the influence of the operating pressure on the permeance. The operating pressure only affects the diffusion in the gas phase. The diffusion in the membrane is not affected by a change in operating pressure. The overall measured permeance increases with lower pressures, as the gas phase diffusion coefficient is higher. The model describes this effect very well.

Performance evaluation To determine how well the model fits the experimental data, parity plots of the permeance Q are shown in

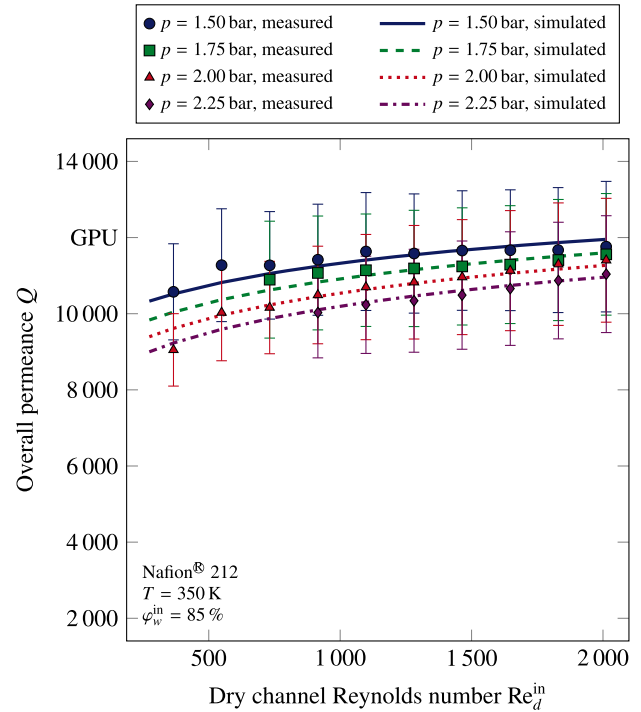


Fig. 6. Permeance Q of Nafion® 212 as a function of the dry channel Reynolds number Re_d^{in} for varying pressure p .

Fig. 7. For Nafion® 212 and 115, the model shows excellent agreement with the experimental data. All points are within the $\pm 20\%$ interval. For Nafion® 212, 77 % of the points are in the range $\pm 10\%$. For Nafion® 211, 49 % of the data are in the $\pm 20\%$ range. All data for Nafion® 211 are in the $\pm 30\%$ range, which is considered good. In the following, we present uncertainties in our model that may contribute to the deviations.

First, the mass transfer through the boundary layer is described with the correlation of Muzychka and Yovanovich [33] assuming analogy of heat and mass transfer and UWT. As mentioned in Section 3.1, neither UWT nor UWF is true in membrane applications. Second, we use a sorption isotherm and a diffusion coefficient to describe the sorption equilibrium and the diffusion in the membrane. Various sorption isotherms and diffusion coefficient models are available in the literature, which differ qualitatively and quantitatively in some cases. All these effects can contribute to deviations between the model and the experimental data.

Overall the results are very good, especially considering that the model is solely based on first principles and empirical equations without any fitting parameters. This suggests that the most important effects contributing to mass transfer are covered well by the model equations.

5.2. Gore composite membrane

The Gore composite membrane consists of a selective PFSA layer with a thickness of $\delta_m < 10 \mu\text{m}$ and supporting layers made from extruded polytetrafluoroethylene (ePTFE) on both sides. Since the selective layer is very thin as compared to the investigated Nafion® membranes, the values of the permeance of the composite membrane are much higher. Fig. 8 shows a parity plot for the Gore composite membrane for various process conditions. The model systematically overestimates the experimental values. The trends within the data are represented correctly; quantitatively, there is a discrepancy. This might indicate an inaccurate model approach for the support layer. Nonetheless, the deviations are mostly less than 30 % without fitting of parameters to the data.

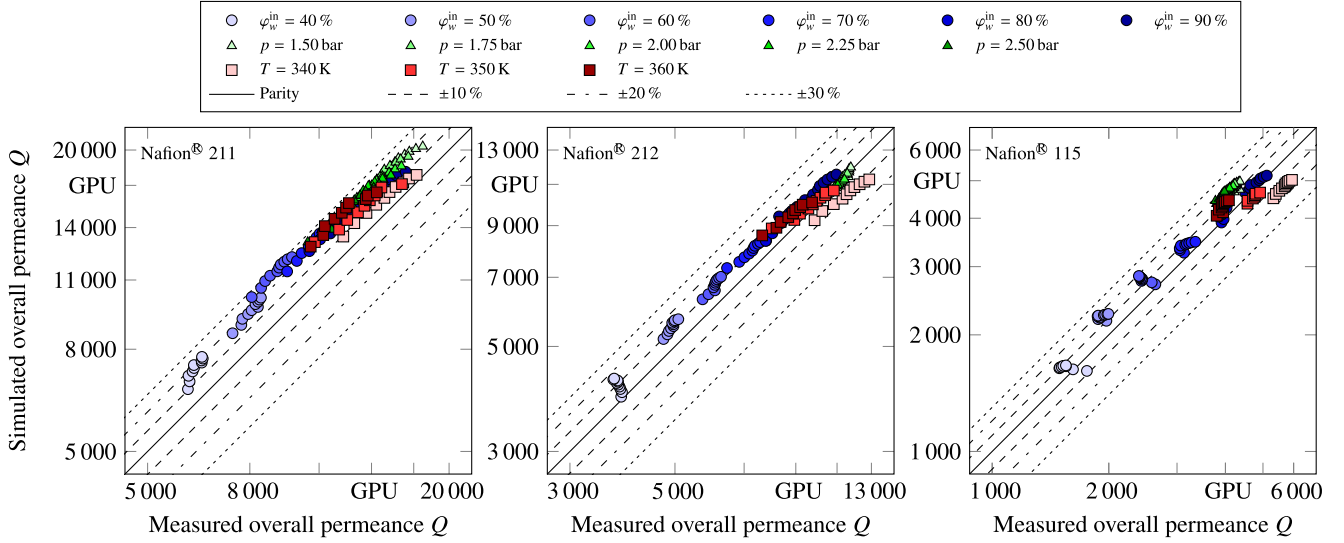


Fig. 7. Parity plots of Nafion® 211, Nafion® 212 and Nafion® 115 from left to right.

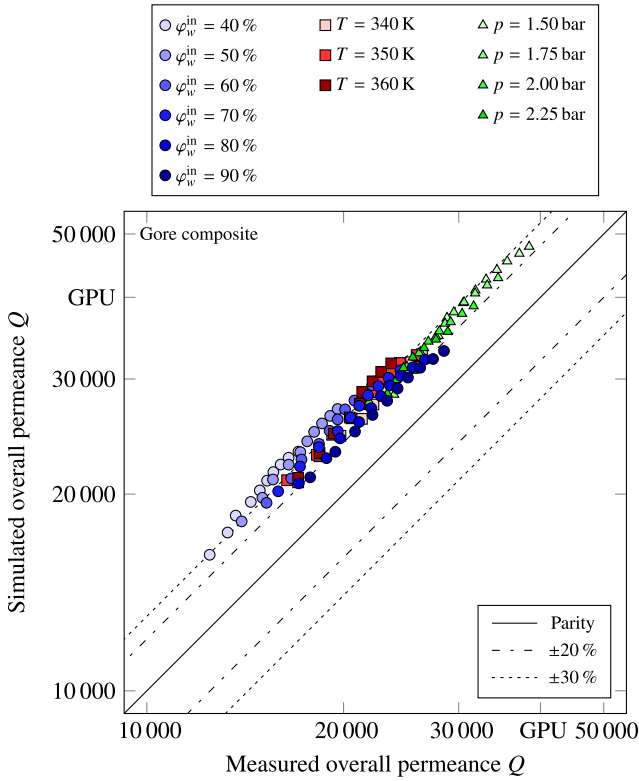


Fig. 8. Parity plot of an experimental Gore composite membrane.

5.3. Mass transfer resistance profiles

The developed model is purely based on first principles and empirical equations. For PFSA membranes like Nafion®, no fitting parameter is needed. Therefore, the model can be utilized to evaluate the influence of each mass transfer resistance. We define the mass transfer resistances as

$$R_i = \frac{\Delta p_{H_2O,i}}{j} \quad (41)$$

Thus, the mass transfer resistance R_w of the wet stream boundary layer is given by

$$R_w = \frac{\bar{R} T_w}{\beta_w \bar{M}_{H_2O}}, \quad (42)$$

the mass transfer resistance R_m for the membrane by

$$R_m = \frac{\delta_m}{\bar{D}_{H_2O,m}} \frac{EW}{\bar{M}_{H_2O} \phi_m} \frac{p_s(T_m)(\phi_{w,m} - \phi_{d,m})}{\Lambda_w - \Lambda_d}, \quad (43)$$

and the mass transfer resistance R_d for the dry stream boundary layer by

$$R_d = \frac{\bar{R} T_d}{\beta_d \bar{M}_{H_2O}}. \quad (44)$$

In Fig. 9, the mass transfer resistances of Nafion® 212 over the relative position $\xi = x/L$ in the flow channel are displayed for dry inlet Reynolds numbers of $Re_d^{\text{in}} = 365$ (left side) and $Re_d^{\text{in}} = 1460$ (right side) and relative humidities of $\phi_w^{\text{in}} = 50\%$ (top) and $\phi_w^{\text{in}} = 85\%$ (bottom). R_{total} is the total mass transfer resistance as a sum of the aforementioned individual resistances. The relative position ξ refers to the dry channel from inlet $\xi = 0$ to outlet $\xi = 1$, while the wet channel is in counter-current.

It is apparent from Fig. 9 that the mass transfer resistances in the boundary layers R_w and R_d are very low at the respective inlets. The resistances grow strongly near the inlets and increase slowly as the flows develop. Since the entry region of the flow is a significant part of the channel length, it is necessary to account for effects of developing flows. At a higher Reynolds number, the boundary layer thickness is thinner, resulting in a lower convective mass transfer resistance on either side over the channel length. The boundary layer resistances at constant Reynolds numbers are approximately equal for both relative humidities.

For each case, the membrane resistance R_m decreases slightly over the length of the dry channel due to a nonlinear dependency of the diffusion coefficient on the water content. At higher relative humidities at the wet side inlet, the diffusion coefficient of the membrane is generally higher across the entire channel length. Therefore, the membrane resistance is significantly lower. Most interesting is the fact that the resistances of the boundary layers and the membrane resistance are in the same order of magnitude for all cases. The contribution of the mass transfer resistance of the boundary layer to the total resistance is most prominent at high humidities and low Reynolds numbers. For such conditions, modeling the boundary layers is crucial to calculate the mass transfer in membrane humidifiers. At low humidities and high Reynolds

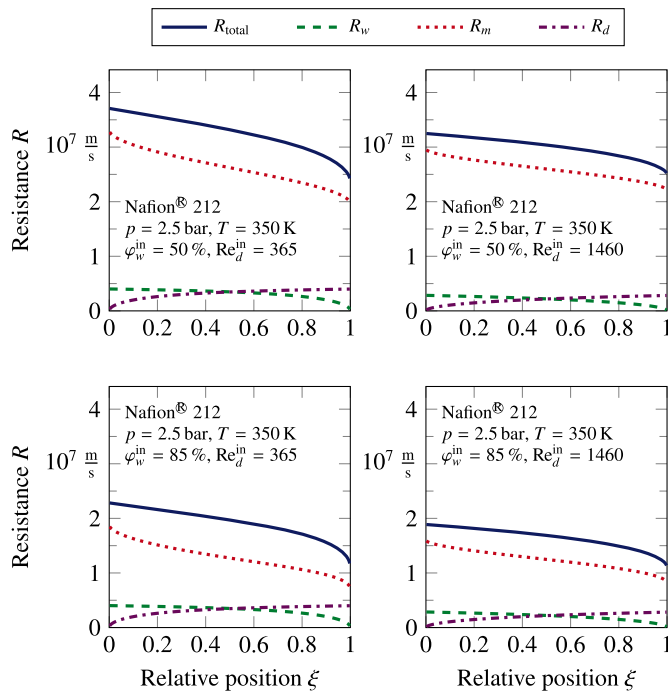


Fig. 9. Mass transfer resistances of Nafion® 212.

numbers, the membrane resistance becomes more dominant. Still, the boundary layer resistance cannot be neglected.

6. Conclusion

In this work, an analytical model for PFSA based membrane humidifiers capturing effects of boundary layers, developing flows and coupled mass and heat transfer was proposed. No fitting parameters are needed to describe the water transport. The model was validated with measurement data of different Nafion® membranes and an experimental composite membrane over a wide range of operating conditions typical for automotive fuel cell applications. The permeance was measured across a range of channel Reynolds numbers with varying wet inlet relative humidity, operating pressure, and temperature. All experiments were performed in a counter-current membrane module.

The measured permeance mainly depends on the wet inlet relative humidity. Higher relative humidities increase the permeance. Increasing the Reynolds number leads to higher values of permeance, whereas an increase of operating temperature or pressure decreases the permeance.

These effects are represented very well by the model. The comparison of measurement data and predicted values of permeance shows excellent agreement for Nafion® 212 and 115 within a deviation of $\pm 20\%$. Since the model does not rely on any fitting parameters, this is remarkable. For Nafion® 211, all data are within a margin of $\pm 30\%$. For the experimental Gore composite membrane, the model overestimates the experimental data up to 40%.

The developed model allows an analysis of the different mass transfer resistances of the water transport in a membrane humidifier. This was showcased for Nafion® 212 at different operating conditions. The analysis shows the necessity for implementing inlet effects and the significant contribution of the boundary layers to the total resistance. We showed that the boundary layer resistances and the membrane resistance are in the same order of magnitude. Therefore, the boundary layer resistance cannot be neglected for channel Reynolds numbers below 2000, which is in the typical range for automotive applications.

CRediT authorship contribution statement

Ladislav Schoenfeld: Methodology, Software, Visualization, Writing – original draft. **Michael Kreitmair:** Methodology, Software, Visualization, Writing – original draft. **Florian Wolfenstetter:** Investigation, Methodology, Software, Writing – review & editing. **Maximilian Neumann:** Methodology, Software, Writing – review & editing. **Harald Klein:** Supervision, Writing – review & editing. **Sebastian Rehfeldt:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

The authors of this paper gratefully acknowledge BMW AG for the financial support. Furthermore, we would like to thank W.L. Gore & Associates for providing the membrane samples.

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